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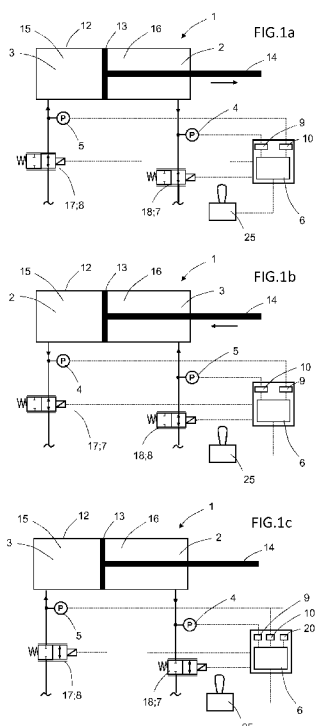
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(54) Title: METHOD AND SYSTEM FOR CONTROLLING THE PRESSURE IN THE CHAMBERS OF A HYDRAULIC CYLINDER



(57) Abstract: The disclosure concerns a method for controlling the pressure in the chambers of a hydraulic cylinder (1) having a meter-out chamber (2) and a meter-in chamber (3). A meter-out valve (7) is connected to the meter-out chamber (2) for regulating the fluid out from the meter-out chamber (2) and a meter-in valve (8) is connected to the meter-in chamber (3) for regulating the fluid into the meter-in chamber (3). The method comprising the steps of sensing the pressure associated with the meter-out chamber (2) with a first pressure sensor (4) which is generating a first pressure signal, sensing the pressure associated with the meter-in chamber (3) with a second pressure sensor (5) which is generating a second pressure signal, processing the first pressure signal in a first signal processing filter (9) into a filtered first pressure signal, wherein in response to the filtered first pressure signal and a filtered or non-filtered second pressure signal delivering a control signal from a control unit (6) to the meter-out valve (7), and wherein the meter-out valve (7) in response to the control signal controls the pressure in the meter-in chamber (3) by regulating the fluid out from the meter-out chamber (2). The disclosure also relates to a corresponding system for controlling the pressure in the chambers of a hydraulic cylinder (1) having a meter-out chamber (2) and a meter-in chamber (3).

METHOD AND SYSTEM FOR CONTROLLING THE PRESSURE IN THE CHAMBERS OF A HYDRAULIC CYLINDER

TECHNICAL FIELD

The present disclosure relates to a method for controlling the pressure in the chambers of a hydraulic cylinder, where the hydraulic cylinder has a meter-out chamber and a meter-in chamber. A meter-out valve is connected to the meter-out chamber for regulating the fluid out from the meter-out chamber, and a meter-in valve is connected to the meter-in chamber for regulating the fluid into the meter-in chamber. The pressure associated with the meter-out chamber is sensed with a first pressure sensor generating a first pressure signal and the pressure associated with the meter-in chamber is sensed with a second pressure sensor generating a second pressure signal. The disclosure further relates to a system for controlling the pressure in the chambers of a hydraulic cylinder.

BACKGROUND

Hydraulic systems are frequently used for work machines or powered construction machines, for example hydraulic excavators, wheel loaders, loading shovels, backhoe shovels, mining equipment, industrial machinery and the like. The hydraulic systems often include one or more actuated components, such as lifting arms, tilting arms, boom assemblies, buckets, and other steering and turning functions or traveling means. A boom assembly may for instance comprise a boom, an arm and a bucket that are pivotally coupled to each other. The actuated components are controlled and operated by hydraulic actuators, which for example can be a hydraulic cylinder or fluid motor that uses hydraulic power to perform the operation.

Traditionally, in such hydraulic systems, a prime mover drives a fixed or variable displacement hydraulic pump for providing hydraulic fluid, such as hydraulic oil, to the hydraulic actuators. Valves are arranged to control the

flow of fluid to the actuators. Today however, the traditional hydraulic systems have often been replaced with electro-hydraulic actuator systems (EHA), where an electric motor is connected to a hydraulic pump for providing fluid to the hydraulic actuator for controlling the mechanical motion
5 of the actuator. The speed and direction of the electric motor and/or the variable displacement hydraulic pump controls the flow of fluid from the pump. Power for the electric motor is received from a power unit, for example a generator or a power storage unit, such as a battery.

For hydraulic cylinders, a cylinder body houses a piston and a piston rod.
10 The piston defines two chambers in the cylinder, one meter-out chamber and one meter-in chamber. The meter-out chamber is the chamber, which hydraulic fluid is flowing out from during operation, and the meter-in chamber is the chamber, which hydraulic fluid is flowing into during operation. Metering valves are regulating the flow into and out from the chambers.
15 Pressurized hydraulic fluid from the hydraulic pump is during powered extension and retraction of the hydraulic cylinder usually applied to the meter-in chamber by a meter-in valve adapted for regulating the flow of pressurized hydraulic fluid into the meter-in chamber. The fluid exhausting from the meter-out chamber flows through a meter-out valve into a return
20 conduit that leads to a system tank, where the meter-out valve is regulating the flow of oil out from the meter-out cylinder.

When operating the hydraulic cylinder a common issue with this type of system configuration with two metering valves is that the dynamic stability of the hydraulic cylinder is low, or that the responsiveness of the hydraulic
25 cylinder is low, or that potentially damaging overpressure may occur in a chamber.

There is thus a need for an improved system and method for controlling the pressure in the chambers of a hydraulic cylinder with two valves controlling the flow of hydraulic fluid into and out from the cylinder chambers, where the
30 system can be operated with high speed and stability.

SUMMARY

One known method is to control the flow into the meter-in cylinder with the meter-in valve, thereby controlling the cylinder speed. The meter-out valve is then free to control the pressure level of the hydraulic cylinder. The target is to control the pressure in the meter-in chamber with the meter-out valve. A control signal to the meter-out valve can be established by a feedback signal of the measured meter-in pressure to a control unit, which is comparing the feedback signal with a reference value. With this method, the dynamic problem between the two cylinder chambers occurs during operation of the hydraulic cylinder. The dynamic stability between the cylinders will become phase shifted in a way that reduces the stability margins of the closed loop control, which means that the controller gain needs to be limited in value. This limitation is for many systems so limiting that required performance when it comes to dynamic stability is not achievable in a satisfactory way, making the system slow and unstable.

An object of the present disclosure is to provide a method and a system where the previously mentioned problems are avoided. This object is at least partly achieved by the features of the independent claims. The dependent claims contain further developments of the method and system.

The disclosure concerns a method for controlling the pressure in the chambers of a hydraulic cylinder having a meter-out chamber and a meter-in chamber, where a meter-out valve is connected to the meter-out chamber for regulating the fluid out from the meter-out chamber. The meter-in valve is connected to the meter-in chamber for regulating the fluid into the meter-in chamber. The method comprises the steps; sensing the pressure associated with the meter-out chamber with a first pressure sensor, which is generating a first pressure signal; sensing the pressure associated with the meter-in chamber with a second pressure sensor, which is generating a second pressure signal; processing the first pressure signal in a first signal processing filter into a filtered first pressure signal; in response to a filtered or

non-filtered first pressure signal and the filtered second pressure signal delivering a control signal from a control unit to the meter-out valve; wherein the meter-out valve in response to the control signal controls the pressure in the meter-in chamber by regulating the fluid out from the meter-out chamber.

- 5 Advantages with these features are that when operating the hydraulic cylinder according to the method involving two metering valves is that the dynamic stability of the hydraulic cylinder is high, and that also the responsiveness of the hydraulic cylinder is high, avoiding potentially damaging overpressure that may occur in the cylinder chamber. Further, with
10 this method the system can be operated with high speed and stability.

According to an aspect of the disclosure, the first signal processing filter allows passing of signals above a first fixed or variable frequency level. By using this approach the meter-in cylinder chamber pressure is controlled statically where dynamically the meter-out pressure is allowed to influence
15 the controller as well to improve stability margins.

According to another aspect of the disclosure, the first signal processing filter comprises a fixed or variable frequency high-pass filter. With a high-pass filter, signals with a frequency higher than a certain cut-off frequency are allowed to pass and signals with frequencies lower than the cut-off frequency
20 are attenuated. The amount of attenuation for each frequency depends on the filter design and the use of a high-pass filter supports the stability of the system.

According to another aspect of the disclosure, the first signal processing filter comprises a high-pass filter with a cut-off frequency in the range of 0.1 – 10
25 Hz, specifically in the range of 0.5 – 5 Hz.

According to a further aspect of the disclosure, the method further comprises the step: processing the second pressure signal in a second signal processing filter into the filtered second pressure signal, where the second signal processing filter allows passing of signals below a second fixed or

variable frequency level. By using a second signal processing filter together with the first signal processing filter, where the second signal processing filter allows passing of signals below a second fixed or variable frequency level, the stability of the system is even further increased because the a more static pressure characteristics of the meter-in cylinder chamber is used for controlling the pressure in the meter-in cylinder chamber.

According to a further aspect of the disclosure, the second signal processing filter additionally allows passing of signals above a third fixed or variable frequency level. Thereby the responsiveness of the hydraulic cylinder may be improved, avoiding potentially damaging overpressure that may occur in the cylinder chamber. Further, with this method the system can be operated with higher speed and stability.

According to a further aspect of the disclosure, the method further comprises the step of processing the second pressure signal in a third signal processing filter into a filtered third pressure signal, where the third signal processing filter allows passing of signals above a third fixed or variable frequency level, and in response to the filtered first, second and third pressure signals delivering a control signal from a control unit to the meter-out valve. Using a separate, individual filter for each input signal to the control unit may enable simplified design of the filters and improved filtering performance.

According to an aspect of the disclosure, the second signal processing filter comprises a fixed or variable frequency low-pass filter, or the second signal processing filter comprises a fixed or variable frequency low-pass filter and a fixed or variable frequency high-pass filter. Through the use of a fixed or variable frequency low-pass filter and/or a fixed or variable frequency high-pass filter, the stability of the system will become high. A low-pass filter is an electronic filter that passes signals with a frequency lower than a certain cut-off frequency and attenuates signals with frequencies higher than the cut-off frequency. The amount of attenuation for each frequency depends on the filter design.

According to another aspect of the disclosure, the second signal processing filter comprises a low-pass filter with a cut-off frequency in the range of 0.1 – 10 Hz, specifically in the range of 0.5 – 5 Hz, or the second signal processing filter comprises a low-pass filter with a cut-off frequency in the range of 0.1 – 10 Hz, specifically in the range of 0.5 – 5 Hz and a high-pass filter with a cut-off frequency in the range of 0.1 – 10 Hz, specifically in the range of 0.5 – 5 Hz.

According to another aspect of the disclosure, the third processing filter comprises a high-pass filter with a cut-off frequency in the range of 0.1 – 10 Hz, specifically in the range of 0.5 – 5 Hz.

According to a further aspect of the disclosure, the control signal from the control unit is based on the sum of the low-pass filtered second pressure signal and the high-pass filtered first pressure signal, or the control signal from the control unit is based on the sum of the low-pass filtered second pressure signal and the high-pass filtered second pressure signal and the high-pass filtered first pressure signal. These system features support the stability and speed of the system.

According to a further aspect of the disclosure, the control signal from the control unit is based on the sum of the non-filtered second pressure signal and the high-pass filtered first pressure signal. Thereby an alternative set-up of the system is provided that in certain situations may provide advantageous performance in terms of stability and speed of the system.

According to an aspect of the disclosure, the first processing filter and/or the second processing filter is a band-pass filter that allows passing of signals between a specific frequency range.

According to an aspect of the disclosure, the method further comprises the steps: measuring the actuating position of the meter-in valve by a valve position sensor, which is generating a valve position signal; supplying the valve position signal to the control unit; and generating a control signal in the

control unit based also the valve position signal and delivering the control signal to the meter-out valve. The valve position sensor is used to further control the system so that the position of the meter-in valve is measured and used as input to the control signal. The meter-out valve is in this way also
5 controlled based on the position of the meter-in valve, which supports a high system speed and stability that gives a high responsiveness of the hydraulic cylinder.

The disclosure further concerns a system for controlling the pressure in the chambers of a hydraulic cylinder having a meter-out chamber and a meter-in
10 chamber. The system comprises; a meter-out valve connected to the meter-out chamber for regulating the fluid out from the meter-out chamber, a meter-in valve connected to the meter-in chamber for regulating the fluid into the meter-in chamber; a first pressure sensor adapted for sensing the pressure associated with the meter-out chamber and for generating a first pressure
15 signal; a second pressure sensor adapted for sensing the pressure associated with the meter-in chamber and for generating a second pressure signal. A first signal processing filter is adapted for processing the first pressure signal into a filtered first pressure signal. A control unit is in response to the filtered first pressure signal and a filtered or non-filtered
20 second pressure signal delivering a control signal to the meter-out valve, where the meter-out valve in response to the control signal controls the pressure in the meter-in chamber by regulating the fluid out from the meter-out chamber.

Advantages with these features are that when operating the hydraulic
25 cylinder according to the system with two metering valves is that the dynamic stability of the hydraulic cylinder is high, and that also the responsiveness of the hydraulic cylinder is high, avoiding potentially damaging overpressure that may occur in the cylinder chamber. Further, the system can be operated with high speed and stability.

According to an aspect of the disclosure, the first signal processing filter allows passing of signals above a first fixed or variable frequency level. By using this approach the meter-in cylinder chamber pressure is controlled statically where dynamically the meter-out pressure is allowed to influence
5 the controller as well to improve stability margins.

According to an aspect of the disclosure, the first signal processing filter comprises a fixed or variable frequency high-pass filter. An advantage with the use of a high-pass filter is that the stability of the system increases when signals with a frequency higher than a certain cut-off frequency are allowed
10 to pass and signals with frequencies lower than the cut-off frequency are attenuated.

According to another aspect of the disclosure, the first signal processing filter comprises a high-pass filter with a cut-off frequency in the range of 0.1 – 10 Hz, specifically in the range of 0.5 – 5 Hz.

15 According to a further aspect of the disclosure, a second signal processing filter is adapted for processing the second pressure signal into the filtered second pressure signal, where the second signal processing filter allows passing of signals below a second fixed or variable frequency level. By using a second signal processing filter together with the first signal processing filter,
20 where the second signal processing filter allows passing of signals below a second fixed or variable frequency, the stability of the system is even further increased.

According to a further aspect of the disclosure, the second signal processing filter additionally allows passing of signals above a third fixed or variable
25 frequency level. Thereby the responsiveness of the hydraulic cylinder will be high, avoiding potentially damaging overpressure that may occur in the cylinder chamber. Further, with this method the system can be operated with higher speed and stability.

According to a further aspect of the disclosure, a third signal processing filter is adapted for processing the second pressure signal into a filtered third pressure signal, where the third signal processing filter allows passing of signals above a third fixed or variable frequency level, and in response to the
5 filtered first, second and third pressure signals delivering a control signal from a control unit to the meter-out valve. Using a separate, individual filter for each input signal to the control unit may enable simplified design of the filters and improved filtering performance.

According to an aspect of the disclosure, the second signal processing filter
10 comprises a fixed or variable frequency low-pass filter, or the second signal processing filter comprises a fixed or variable frequency low-pass filter and a fixed or variable frequency high-pass filter. Through the use of a fixed or variable frequency low-pass filter and/or a fixed or variable frequency high-pass filter, the stability of the system will become high.

According to another aspect of the disclosure, the second signal processing filter comprises a low-pass filter with a cut-off frequency in the range of 0.1 – 10 Hz, specifically in the range of 0.5 – 5 Hz, or the second signal processing filter comprises a low-pass filter with a cut-off frequency in the range of 0.1 – 10 Hz, specifically in the range of 0.5 – 5 Hz and a high-pass filter with a cut-
20 off frequency in the range of 0.1 – 10 Hz, specifically in the range of 0.5 – 5 Hz.

According to another aspect of the disclosure, the third processing filter comprises a high-pass filter with a cut-off frequency in the range of 0.1 – 10 Hz, specifically in the range of 0.5 – 5 Hz.

According to an aspect of the disclosure, the control signal from the control unit is based on the sum of the low-pass filtered second pressure signal and the high-pass filtered first pressure signal, or the control signal from the control unit is based on the sum of the low-pass filtered second pressure signal and the high-pass filtered second pressure signal and the high-pass
25

filtered first pressure signal. These system features support the stability and speed of the system.

According to an aspect of the disclosure, the control signal from the control unit is based on the sum of the non-filtered second pressure signal and the high-pass filtered first pressure signal. Thereby an alternative set-up of the system is provided that in certain situations may provide advantageous performance in terms of stability and speed of the system.

According to another aspect of the disclosure, the first processing filter and/or the second processing filter is a band-pass filter that allows passing of signals between a specific frequency range.

According to an aspect of the disclosure, a valve position sensor is adapted for measuring the actuating position of the meter-in valve and for generating a valve position signal, and the system is adapted to supply the valve position signal to the control unit, and the control unit is adapted to generate a control signal based also on the valve position signal and to deliver the control signal to the meter-out valve. The valve position sensor further controls the system so that the position of the meter-in valve is measured and used as input to the control signal. The meter-out valve is thus controlled with input based on the position of the meter-in valve, which supports a high system speed and stability that gives a high responsiveness of the hydraulic cylinder.

BRIEF DESCRIPTION OF DRAWINGS

The disclosure will be described in greater detail in the following, with reference to the attached drawings, in which

Fig. 1a-b show schematically, a system for controlling the pressure in the chambers of a hydraulic cylinder according to the disclosure,

Fig. 1c show schematically, an alternative embodiment of the system of fig. 1a,

Fig. 2-4 show graphs schematically illustrating example behaviors of a system for controlling the pressure in the chambers of a hydraulic cylinder according to the disclosure.

DESCRIPTION OF EXAMPLE EMBODIMENTS

- 5 Various aspects of the disclosure will hereinafter be described in conjunction with the appended drawings to illustrate and not to limit the disclosure, wherein like designations denote like elements, and variations of the described aspects are not restricted to the specifically shown embodiments, but are applicable on other variations of the disclosure.
- 10 Figures 1a and 1b schematically show a hydraulic system with a hydraulic cylinder 1 that uses hydraulic power to perform an operation. Hydraulic cylinders are often used to actuate loads in systems for work machines or powered construction machines, such as hydraulic excavators, wheel loaders, loading shovels, backhoe shovels, mining equipment, industrial
- 15 machinery and the like. The hydraulic cylinder may impart a force while either extending or retracting which in turn moves the load respectively. Hydraulic systems often include one or more hydraulic cylinders.

The hydraulic cylinder 1 has a cylinder body 12 that is housing a piston 13 and a piston rod 14. The piston rod 14 is applying a desired force upon the

20 actuated component. The piston 13 defines two chambers in the cylinder, a first cylinder chamber 15 and a second cylinder chamber 16. During operation of the hydraulic cylinder, either the first cylinder chamber 15 or the second cylinder chamber 16, depending on the operational direction of the hydraulic cylinder, may be filled with hydraulic fluid from a hydraulic pump.

25 The other cylinder chamber is during operation emptied and the hydraulic fluid may flow out from the cylinder chamber to a reservoir. The first cylinder chamber 1 and the second cylinder chamber 2 have variable volumes and the volumes are depending on the position of the piston 13, which moves during operation of the hydraulic cylinder 1. During a filling sequence the

30 cylinder chambers are expanding and thus increasing in volume and when

being emptied, the cylinder chambers are contracting and decreasing in volume.

The cylinder chamber that is filled with hydraulic fluid is referred to as the meter-in chamber 3 and the cylinder chamber that is being emptied is referred to as the meter-out chamber 2. Both the first cylinder chamber 15 and the second cylinder chamber 16 are thus acting as meter-in and meter-out chambers depending on the operational direction of the hydraulic cylinder. The meter-out chamber 2 is the chamber, which hydraulic fluid is flowing out from during operation, and the meter-in chamber 2 is the chamber, which hydraulic fluid is flowing into during operation.

In the hydraulic system, a prime mover may drive a hydraulic pump for providing the hydraulic fluid, such as hydraulic oil, to the hydraulic cylinder 1. Also, the hydraulic system may be of the electro-hydraulic actuator type, where an electric motor is connected to the hydraulic pump for providing fluid to the hydraulic cylinder 1 for controlling the mechanical motion of the hydraulic cylinder 1. The speed and direction of the electric motor may control the flow of fluid to the actuator. Power for the electric motor may be received from a power unit, for example a generator or a power storage unit, such as a battery.

In the hydraulic system, individual proportional control valves may be arranged to regulate the flow of fluid into and out from the cylinder chambers. The proportional control valves may typically be electrically operated solenoid valves suitable for being individually operated by an electronic control unit. A first proportional control valve 17 is controlling the hydraulic fluid into and out from the first cylinder chamber 15 and a second proportional control valve 18 is controlling the hydraulic fluid into and out from the second cylinder chamber 16. The first and second control valves 17, 18 can be of any kind, i.e. flow controlling, throttling or pressure controlling. The control valve that is controlling the hydraulic fluid out from the meter-out chamber 2 is referred to as the meter-out valve 7 and the valve that is controlling the fluid into the

meter-in chamber 3 is referred to as the meter-in valve 8. Both the first control valve 17 and the second control valve 18 are thus acting as meter-in and meter-out valves depending on the operational direction of the hydraulic cylinder.

- 5 Alternatively, the valve system for controlling the motion of the hydraulic cylinder 1 may comprise four individual proportional control valves, as illustrated in fig. 1 of WO 2012/161628A1 which is incorporated herein. In such a valve system a first control valve is arranged to act as meter-in valve of the first cylinder chamber 15, a second control valve is arranged to act as meter-out valve of the first cylinder chamber 15, a third control valve is arranged to act as meter-in valve of the second cylinder chamber 16, and a fourth control valve is arranged to act as meter-out valve of the second cylinder chamber 16, wherein each of said four control valves is controlled individually.
- 10
- 15 In fig. 1a, pressurized hydraulic fluid from the hydraulic pump may during powered extension and retraction of the hydraulic cylinder 1 be applied to the meter-in chamber 3 via the meter-in valve 8, which is adapted for regulating the flow of pressurized hydraulic fluid into the meter-in chamber 3. The fluid exhausting from the meter-out chamber 2 flows through the meter-out valve 7 into a return conduit that for example may lead to a system tank, where the meter-out valve 7 is adapted for regulating the flow of oil out from the meter-out cylinder 2.
- 20

The system for controlling the pressure in the chambers of a hydraulic cylinder 1, comprises the meter-out chamber 2, the meter-in chamber 3, the meter-out valve 7 connected to the meter-out chamber 2 for regulating the fluid out from the meter-out chamber 2, and the meter-in valve 8 connected to the meter-in chamber 3 for regulating the fluid into the meter-in chamber 3. By means of the proportional control character of the meter-in and meter-out valves 8, 7 the piston speed and cylinder chamber pressures can be accurately controlled. As shown in figures 1a and 1b, the control

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system further comprises a first pressure sensor 4, which is adapted for sensing the pressure associated with the meter-out chamber 2, and a second pressure sensor 5 adapted for sensing the pressure associated with the meter-in chamber 3.

- 5 The first pressure sensor 4 and the second pressure sensor 5 may be of any conventional pressure sensor type suitable for measuring the pressure in the respective cylinder chambers. The second pressure sensor 5 may be located at the hydraulic cylinder 1, at the meter-in valve 8, or anywhere along the fluid pipe connecting the meter-in valve 8 with the hydraulic cylinder 1. The
- 10 same applies to the first pressure sensor 4 correspondingly. The first pressure sensor 4 is generating a first pressure signal and the second pressure sensor 5 is generating a second pressure signal.

The control system has a control unit 6, for example an electronic control unit (ECU) that may be in the form of a central processing unit (CPU) or similar

15 device, which is controlling the different system components. The control unit 6 is connected to the first pressure sensor 4 and the second pressure sensor 5 so that the first pressure signal and the second pressure signal is received and processed by the control unit 6. Further the control unit 6 is connected to the meter-out valve 7 and the meter-in valve 8 so that the respective valves

20 can be controlled by the control unit 6 in response to various input signals, such as a user actuated joystick 25 and sensor input signals. The controlling of the valves may be achieved by output signals sent to the valves and valve regulators attached on the valves may in response to the control signals set the valves in a specific position. The valves may for example either be set in

25 closed position, a fully open position, or in any desired open position between the closed position and the fully open position, in order to control the flow of hydraulic fluid out from or into the cylinder chambers.

As described above, when operating a hydraulic cylinder, a common issue with this type of system configuration with two metering valves is that the

30 dynamic stability of the hydraulic cylinder is not optimal, when using the

meter-in pressure value as feedback signal in a closed-loop controller configured to control the pressure of the meter-in chamber by controlling the flow through the meter-out valve. With this method, a dynamic phase-shift between the two cylinder chambers occurs during operation of the hydraulic cylinder due to the inertia of the system connected to the piston rod 14.

Fig. 2 schematically illustrates an example behavior of a relatively high controller gain hydraulic control system comprising a hydraulic cylinder with a piston rod connected to a substantial load, such as a filled bucket on a wheel-loader, and where the meter-in pressure value is used as feedback signal in a closed-loop controller configured to control the pressure of the meter-in chamber by regulating the fluid through the meter-out valve. At time t_1 the controller receives instruction to commence extension of the piston at a certain constant speed. The controller may thereupon adjust the meter-in valve to correspond to the target piston speed, and collect a suitable target nominal pressure P_1 for the meter-in valve from a memory. Since the pressure value 21 in the meter-in chamber is lower than P_1 at time t_1 the controller reduces the flow passage area of the meter-out valve. As a result, the pressure 21 in the meter-in chamber increases. When the pressure 21 in the meter-in valve reaches nominal pressure P_1 , the nominal-actual value difference is zero and the controller has opened the flow passage area of the meter-out valve again. However, the significant load of the system connected to the piston rod results in a significant inertia of the system, such that the piston accelerates relatively slowly toward the target speed. Consequently, the pressure 21 in the meter-in chamber increases beyond the nominal pressure P_1 , and the controller opens the flow passage area of the meter-out valve even further for reducing the pressure 21 in the meter-in chamber. At time t_2 the increased flow passage area of the meter-out valve has finally enabled the meter-in pressure 21 to arrive the nominal pressure P_1 again, the nominal-actual value difference is zero and the controller has opened the flow passage area of the meter-out valve again. However, the large inertia of the system again results in an undershoot of the meter-in pressure 21, and

sequence repeats oppositely, thereby resulting in an instable extension of the piston.

The dynamic stability margins of the closed loop control is low due to the large inertia of the moving system, which means that the controller gain
5 needs to be limited in value. This limitation is for many systems so limiting that required performance when it comes to system responsiveness and agility is not achievable in a satisfactory way.

When operating a hydraulic cylinder system with two metering valves, such as the meter-out valve 7 and the meter-in valve 8, it is thus important to
10 balance the controlling of the system to achieve a dynamic stability between the two cylinder chambers. The flow of hydraulic fluid into the meter-in chamber 3 and the flow out from the meter-out chamber 2 must be balanced in a way so that instability is avoided when operating the hydraulic cylinder and controlling the cylinder speed.

15 According to the disclosure, dynamic stability of the hydraulic system is improved by controlling the valves with input from the first pressure signal generated by the first pressure sensor 4 and the second pressure signal generated by the second pressure sensor 5.

Fig. 3 schematically illustrates the example behavior of fig. 2 but also
20 including measured meter-out pressure 22. Fig. 3 illustrates schematically the dynamic phase shift in pressure between the meter-in pressure 21 and meter-out pressure 22 that may occur depending on the system characteristics. In the example embodiment of fig. 3 there is a nearly 180 degrees pressure phase shift between the meter-in and meter-out pressures
25 21, 22. The degree of phase-shift is however machine specific and load specific.

By also incorporating the high-frequency part of the pressure value 22 of the meter-out chamber 2, the pressure phase shift can be taken into account when controlling the hydraulic cylinder. In a system where the meter-in

pressure is controlled, the method with using both a high-pass filtered first pressure signal and filtered or non-filtered second pressure signal as input to the controlling of the meter-out valve 7 will provide dynamic stability when also using one or more signal processing filters to achieve complete or partial suppression of some aspect of the signal from the meter-in and/or meter-out pressure sensors.

According to the disclosure, a first signal processing filter 9 is adapted for processing the first pressure signal from the first pressure sensor 4. The first signal processing filter 9, which may be integrated in the control unit 6, is used to remove unwanted components or features of the first pressure signal. A signal processing filter is a circuit that can be designed to modify, reshape or reject all unwanted frequencies of an electrical signal and accept or pass only those signals that are wanted. In other words, the signal processing filter may be designed to remove unwanted signals. Signal processing filters are commonly used to achieve complete or partial suppression of some aspect of the signal, such as for example removing some of the frequencies of the signal but not others in order to suppress interfering signals and reduce background noise. The first signal processing filter 9 is receiving and processing the first pressure signal into a filtered first pressure signal. As an alternative, the first signal processing filter may be arranged in a separate processing unit, which is connected to the control unit 6.

The second pressure signal from the second pressure sensor 5 may in the control system be used as a non-filtered pressure signal or as filtered second pressure signal by the control unit 6.

The meter-out valve 7 can be used for controlling the pressure in the hydraulic cylinder 1. The meter-in valve 8 is controlling the flow of hydraulic fluid from the hydraulic pump into the meter-in chamber 3 and may be used for controlling the cylinder speed. The meter-out valve 7 may then be used to control the pressure level of the hydraulic cylinder 1, both for the meter-out

chamber 2 and the meter-in chamber 2. The pressure level in the meter-out chamber 2 and the meter-in chamber 3 can be varied with the meter-out valve 7 by regulating the flow of hydraulic fluid out from the meter-out chamber 2. By controlling the pressure in the meter-in chamber 3 in a controlled way with the meter-out valve 7, an efficient operation of the hydraulic cylinder with dynamic stability can be achieved.

A system with high dynamic stability and speed is achieved by using an output signal from the control unit 6 to the meter-out valve 7, which output signal is based on both the filtered first pressure signal and the filtered or non-filtered second pressure signal. The control unit 6 thus delivers a control signal to the meter-out valve 7 in response to the filtered first pressure signal and the filtered or non-filtered second pressure signal. The meter-out valve 7 is then in response to the control signal controlling the pressure in the meter-in chamber 3 by regulating the flow out from the meter-out chamber 2.

The control unit 6 samples the pressure signals from the first pressure sensor 4 and the second pressure sensor 5 with a certain sampling frequency. The control unit 6 is then processing the pressure signals based on a system algorithm suitable for the hydraulic system into the control signal, which is used for regulating the meter-out valve 7.

The meter-out valve 7 controls the pressure in the meter-in chamber 3 in response to the control signal, by regulating the flow out from the meter-out chamber 2. The first signal processing filter 9 is in a first embodiment designed to allow passing of signals above a first fixed or variable frequency level. With this setup the dynamic pressure variations of the meter-out chamber 2 of the hydraulic system are taken in account when controlling the system. The first pressure signal from the first pressure sensor 4 associated with the meter-out chamber 2 is after being processed in the first processing filter 9 controlling the dynamic frequency variations of the system. Suitable frequency levels that are passing the first signal processing filter 9 may be

frequencies that are typically a bit lower than the system's resonance frequency and above.

The first processing filter 9 may for example comprise a fixed or variable frequency high-pass filter. A high-pass filter is an electronic filter that passes
5 signals with a frequency higher than a certain cut-off frequency and attenuates signals with frequencies lower than the cut-off frequency. The amount of attenuation for each frequency depends on the filter design. The high-pass filter may have a cut-off frequency in the range of 0.1 – 10 Hz, specifically in the range of 0.5 – 5 Hz, designed to allow passing of signals
10 above the first fixed or variable frequency level, in order to efficiently achieve the desired system performance.

The first processing filter 9 may comprise a static frequency high-pass filter with a fixed cut-off frequency. This solution may be desirable for enabling a more robust design with less risk for malfunction. Alternatively, the first
15 processing filter 9 may comprise a dynamic high-pass filter with a variable cut-off frequency. Such a filter may then adjust its cut-off frequency to various specific situations, such as the specific piston chamber pressure or load of the application, the specific piston position, ambient conditions, etc. This solution may deliver improved performance but requires generally a more
20 complex implementation.

Fig.4 schematically illustrates an example result of this control strategy on the embodiment of figure 2. The input signal to the controller here comprises the sum of the non-filtered second pressure signal the high-pass filtered first pressure signal. As a result the previous undershoot and instability of the
25 meter-in pressure is practically eliminated and the hydraulic system is stable while still having a quick step response due to the relatively high controller gain.

According to an example embodiment, a second signal processing filter 10, adapted for processing the second pressure signal into the filtered second

pressure signal may be used in combination with the first signal processing filter 9. The second signal processing filter 10 may be designed to allow passing of signals below a second fixed or variable frequency level.

The second processing filter 10 may for example comprise a fixed or variable frequency low-pass filter. The input signal to the controller could then comprise the sum of the low frequency part of the second pressure signal and the high frequency part of the first pressure signal. A low-pass filter is an electronic filter that passes signals with a frequency lower than a certain cut-off frequency and attenuates signals with frequencies higher than the cut-off frequency. The amount of attenuation for each frequency depends on the filter design. The low-pass filter may have a cut-off frequency in the range of 0.1 – 10 Hz, specifically in the range of 0.5 – 5 Hz, to allow passing of signals below the second fixed or variable frequency level, in order to achieve the desired system performance. With this setup the static pressure value of the meter-in chamber 3 of the hydraulic system are taken in account when controlling the system. The second pressure signal from the second pressure sensor 5 associated with the meter-in chamber 3 is after being processed in the second processing filter 10 controlling the more static frequency of the system. Suitable frequency levels that are passing the second signal processing filter 9 may be frequencies that are well below the system's resonance frequency.

In a system where the input signal to the controller comprises the sum of the low frequency part of the second pressure signal and the high frequency part of the first pressure signal, the cut-off frequency of the first and second filters 9, 10 may be the same, or the cut-off frequency of the first filter may be set higher than the cut-off frequency of the second filter 10, or the cut-off frequency of the first filter may be set lower than the cut-off frequency of the second filter 10.

According to still an example embodiment, the input signal to the controller may comprise the sum of the low-pass filtered second pressure signal and

the high-pass filtered first pressure signal. This control strategy may be advantageous in certain situations and applications.

According to still an example embodiment, the second signal processing filter 10, may be designed to allow passing of signals below the second fixed or variable frequency level as described above and also allow passing of signals above the third fixed or variable frequency level. The second processing filter 10 may then for example comprise a fixed or variable frequency low-pass filter in combination with a fixed or variable frequency high-pass filter. The input signal to the controller could then be the sum of the low frequency second pressure signal and the high frequency second pressure signal and the high frequency first pressure signal. With this combined setup, the low-pass filter of the second processing filter 10 may have a cut-off frequency in the range of 0.1 – 10 Hz, specifically in the range of 0.5 – 5 Hz, to allow passing of signals below the second fixed or variable frequency level, in order to achieve the desired system performance, and the high-pass filter of the second processing filter 10 may have a cut-off frequency in the range of 0.1 – 10 Hz, specifically in the range of 0.5 – 5 Hz, to allow passing of signals above the third fixed or variable frequency level, in order to achieve the desired system performance. However, the cut-off frequency of the low-pass filter of the second processing filter 10 is lower than the cut-off frequency of the high-pass filter of the second processing filter 10.

With this setup the static pressure value and the dynamic pressure variations of the meter-in chamber 3 of the hydraulic system are taken in account when controlling the system to even further improve the speed and stability margins of the hydraulic system. The control signal from the control unit 6 is then based on the sum of the filtered second pressure signal and the filtered first pressure signal.

Figure 1c schematically illustrates still a further alternative embodiment. Here the second pressure signal may be supplied to a third signal processing filter

20, which is adapted to process the second pressure signal into a filtered third pressure signal. The third signal processing filter 20 allows passing of signals above a third fixed or variable frequency level. Thereby, the control unit can be arranged to supply a control signal to the meter-out valve in response to the filtered first, second and third pressure signals. The third processing filter 20 may then for example comprise a fixed or variable frequency high-pass filter. The high-pass filter may have a cut-off frequency in the range of 0.1 – 10 Hz, specifically in the range of 0.5 – 5 Hz, to allow passing of signals above the third fixed or variable frequency level, in order to achieve the desired system performance. However, the cut-off frequency of the second processing filter 10 is lower than the cut-off frequency of the third processing filter 20.

In yet another alternative embodiment, the first processing filter 9 and/or the second processing filter 10 are band-pass filters allowing passing of signals between specific frequency ranges. A band pass filter allows signals falling within a certain frequency band setup between two points to pass through while blocking both the lower and higher frequencies on either side of this frequency band.

The hydraulic system may further comprise a valve position sensor, which is adapted for measuring the actuating position of the meter-in valve 8 and for generating a valve position signal. The control unit 6 may then use the valve position signal as an input value to the control signal delivered to the meter-out valve 7. In this way the system may be controlled with additional input from the valve position sensor to further, which may increase the speed and stability of the system, since the position of the meter-in valve 8 and thus the flow into the meter-in chamber 3 is taken into consideration.

The above-mentioned processing filters are preferably dynamic filters that take the current operating conditions into account, such as the current load condition, current hydraulic fluid temperature condition, current ambient temperature condition, or other operating conditions that may affect the

dynamic pressure phase-shift of the meter-in and meter-out chambers. The dynamic filters may thus continuously adapt their filter parameters, such as cut-off frequency, etc. to the current operating conditions.

It will be appreciated that the above description is merely exemplary in nature and is not intended to limit the present disclosure, its application or uses. While specific examples have been described in the specification and illustrated in the drawings, it will be understood by those of ordinary skill in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the present disclosure as defined in the claims. Furthermore, modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular examples illustrated by the drawings and described in the specification as the best mode presently contemplated for carrying out the teachings of the present disclosure, but that the scope of the present disclosure will include any embodiments falling within the foregoing description and the appended claims. Reference signs mentioned in the claims should not be seen as limiting the extent of the matter protected by the claims, and their sole function is to make claims easier to understand.

REFERENCE SIGNS

- | | |
|-------|--------------------------------|
| 1: | Hydraulic cylinder |
| 2: | Meter-out chamber |
| 3: | Meter-in chamber |
| 25 4: | First pressure sensor |
| 5: | Second pressure sensor |
| 6: | Control unit |
| 7: | Meter-out valve |
| 8: | Meter-in valve |
| 30 9: | First signal processing filter |

- 10: Second signal processing filter
 - 12: Cylinder body
 - 13: Piston
 - 14: Piston rod
 - 5 15: First cylinder chamber
 - 16: Second cylinder chamber
 - 17: First control valve
 - 18: Second control valve
 - 20: Third signal processing filter
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CLAIMS

1. A method for controlling the pressure in the chambers of a hydraulic cylinder (1) having a meter-out chamber (2) and a meter-in chamber (3),
5 where a meter-out valve (7) is connected to the meter-out chamber (2) for regulating the fluid out from the meter-out chamber (2), and a meter-in valve (8) is connected to the meter-in chamber (3) for regulating the fluid into the meter-in chamber (3), the method comprising the steps:
sensing the pressure associated with the meter-out chamber (2)
10 with a first pressure sensor (4), which is generating a first pressure signal;
sensing the pressure associated with the meter-in chamber (3) with a second pressure sensor (5), which is generating a second pressure signal;
15 processing the first pressure signal in a first signal processing filter (9) into a filtered first pressure signal;
in response to the filtered first pressure signal and a filtered or non-filtered second pressure signal delivering a control signal from a control unit (6) to the meter-out valve (7);
20 wherein the meter-out valve (7) in response to the control signal controls the pressure in the meter-in chamber (3) by regulating the fluid out from the meter-out chamber (2).
2. A method according to claim 1, wherein the first signal processing filter (9) allows passing of signals above a first fixed or variable frequency level
25
3. A method according to any of the preceding claims 1 to 2, wherein the first signal processing filter (9) comprises a fixed or variable frequency
30 high-pass filter.

4. A method according to any of the preceding claims 1 to 3, wherein the first signal processing filter (9) comprises a high-pass filter with a cut-off frequency in the range of 0.1 – 10 Hz, specifically in the range of 0.5 – 5 Hz.

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5. A method according to any of claims 1 to 4, wherein the method further comprises the step:

processing the second pressure signal in a second signal processing filter (10) into a filtered second pressure signal, where the second signal processing filter (10) allows passing of signals below a second fixed or variable frequency level.

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6. A method according to claim 5, wherein the second signal processing filter (10) additionally allows passing of signals above a third fixed or variable frequency level.

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7. A method according to any of claims 1 to 5, wherein the method further comprises the step:

processing the second pressure signal in a third signal processing filter (20) into a filtered third pressure signal, where the third signal processing filter (20) allows passing of signals above a third fixed or variable frequency level, and in response to the filtered first, second and third pressure signals delivering a control signal from a control unit (6) to the meter-out valve (7).

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8. A method according to any of claims 1 to 7,

wherein the second signal processing filter (10) comprises a fixed or variable frequency low-pass filter, or

wherein the second signal processing filter (10) comprises a fixed or variable frequency low-pass filter and a fixed or variable frequency high-pass filter.

30

9. A method according to claim 8,
wherein the second signal processing filter (10) comprises a low-pass filter with a cut-off frequency in the range of 0.1 – 10 Hz, specifically in the range of 0.5 – 5 Hz, or
- 5 wherein the second signal processing filter (10) comprises a low-pass filter with a cut-off frequency in the range of 0.1 – 10 Hz, specifically in the range of 0.5 – 5 Hz, and a high-pass filter with a cut-off frequency in the range of 0.1 – 10 Hz, specifically in the range of 0.5 – 5 Hz.
- 10 10. A method according to claim 7, wherein the third processing filter (20) comprises a high-pass filter with a cut-off frequency in the range of 0.1 – 10 Hz, specifically in the range of 0.5 – 5 Hz.
11. A method according to any of previous claims 8 to 10,
- 15 wherein the control signal from the control unit (6) is based on the sum of the low-pass filtered second pressure signal and the high-pass filtered first pressure signal, or
- wherein the control signal from the control unit (6) is based on the sum of the low-pass filtered second pressure signal and the high-pass filtered second pressure signal and the high-pass filtered first pressure signal.
- 20 wherein the control signal from the control unit (6) is based on the sum of the non-filtered second pressure signal and the high-pass filtered first pressure signal.
12. A method according to any of previous claims 3 to 10, wherein the control signal from the control unit (6) is based on the sum of the non-
- 25 filtered second pressure signal and the high-pass filtered first pressure signal.
13. A method according to any of claims 1 to 12, wherein the method further comprises the steps:
- 30 measuring the actuating position of the meter-in valve (8) by a valve position sensor, which is generating a valve position signal;
 supplying the valve position signal to the control unit (6); and

generating a control signal in the control unit (6) based also on the valve position signal and delivering the control signal to the meter-out valve (7).

- 5 14. A system for controlling the pressure in the chambers of a hydraulic cylinder (1) having a meter-out chamber (2) and a meter-in chamber (3), wherein the system comprises:

a meter-out valve (7) connected to the meter-out chamber (2) for regulating the fluid out from the meter-out chamber (2),

- 10 a meter-in valve (8) connected to the meter-in chamber (3) for regulating the fluid into the meter-in chamber (3);

a first pressure sensor (4) adapted for sensing the pressure associated with the meter-out chamber (2) and for generating a first pressure signal;

- 15 a second pressure sensor (5) adapted for sensing the pressure associated with the meter-in chamber (3) and for generating a second pressure signal;

a first signal processing filter (9) adapted for processing the first pressure signal into a filtered first pressure signal;

- 20 wherein a control unit (6) in response to the filtered first pressure signal and a filtered or non-filtered second pressure signal delivers a control signal to the meter-out valve (7), where the meter-out valve (7) in response to the control signal controls the pressure in the meter-in chamber (3) by regulating the fluid out from the meter-out chamber (2).

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15. A system according to claim 14, wherein the first signal processing filter (9) allows passing of signals above a first fixed or variable frequency level

- 30 16. A system according to any of claims 14 to 15, wherein the first signal processing filter (9) comprises a fixed or variable frequency high-pass filter.

17. A system according to any of claims 14 to 16, wherein the first signal processing filter (9) comprises a high-pass filter with a cut-off frequency in the range of 0.1 – 10 Hz, specifically in the range of 0.5 – 5 Hz.

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18. A system according to any of claims 14 to 17, wherein a second signal processing filter (10) is adapted for processing the second pressure signal into the filtered second pressure signal, and wherein the second signal processing filter (10) allows passing of signals below a second fixed or variable frequency level.

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19. A system according to any of claims 14 to 18, wherein the second signal processing filter (10) additionally allows passing of signals above a third fixed or variable frequency level.

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20. A system according to any of claims 14 to 18, wherein a third signal processing filter (20) is adapted for processing the second pressure signal into a filtered third pressure signal, and wherein the third signal processing filter (20) allows passing of signals above a third fixed or variable frequency level, and in response to the filtered first, second and third pressure signals delivering a control signal from a control unit (6) to the meter-out valve (7).

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21. A system according to any of claims 14 to 20,

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wherein the second signal processing filter (10) comprises a fixed or variable frequency low-pass filter, or

wherein the second signal processing filter (10) comprises a fixed or variable frequency low-pass filter and a fixed or variable frequency high-pass filter.

30

22. A system according to claim 21,

wherein the second signal processing filter (10) comprises a low-pass filter with a cut-off frequency in the range of 0.1 – 10 Hz, specifically in the range of 0.5 – 5 Hz, or

5 wherein the second signal processing filter (10) comprises a low-pass filter with a cut-off frequency in the range of 0.1 – 10 Hz, specifically in the range of 0.5 – 5 Hz and a high-pass filter with a cut-off frequency in the range of 0.1 – 10 Hz, specifically in the range of 0.5 – 5 Hz.

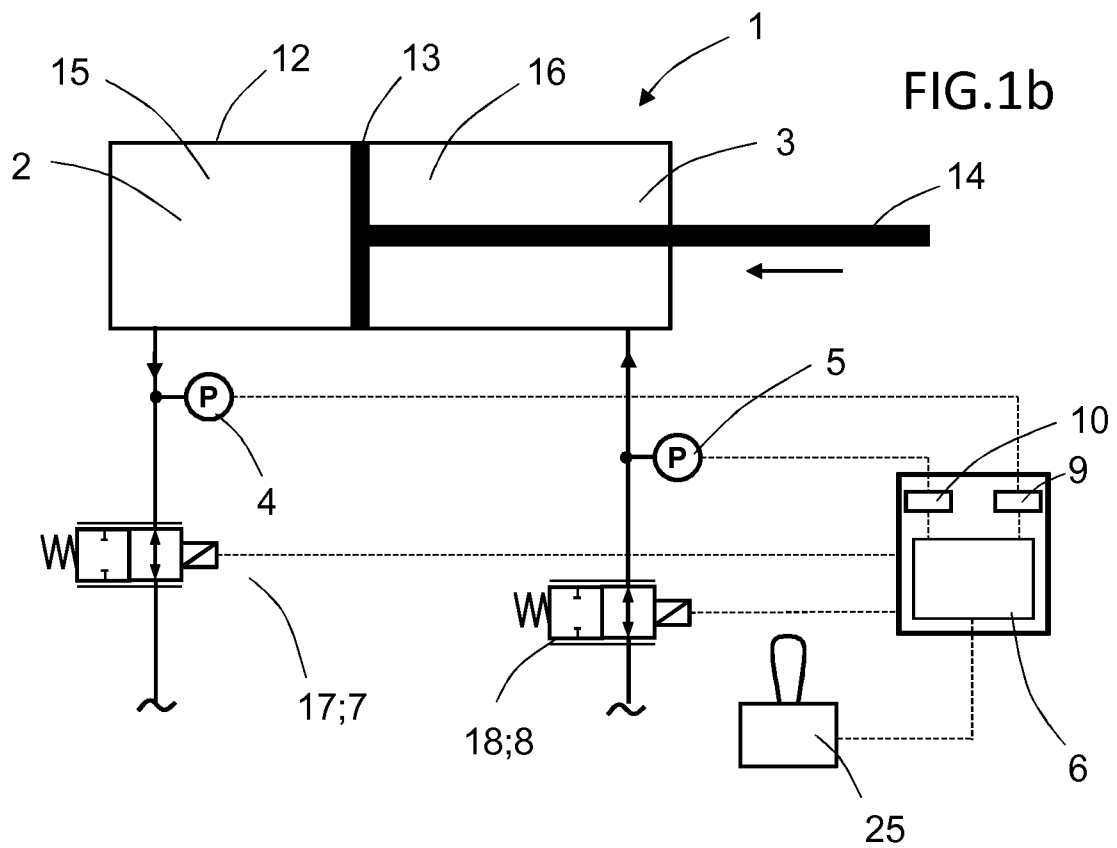
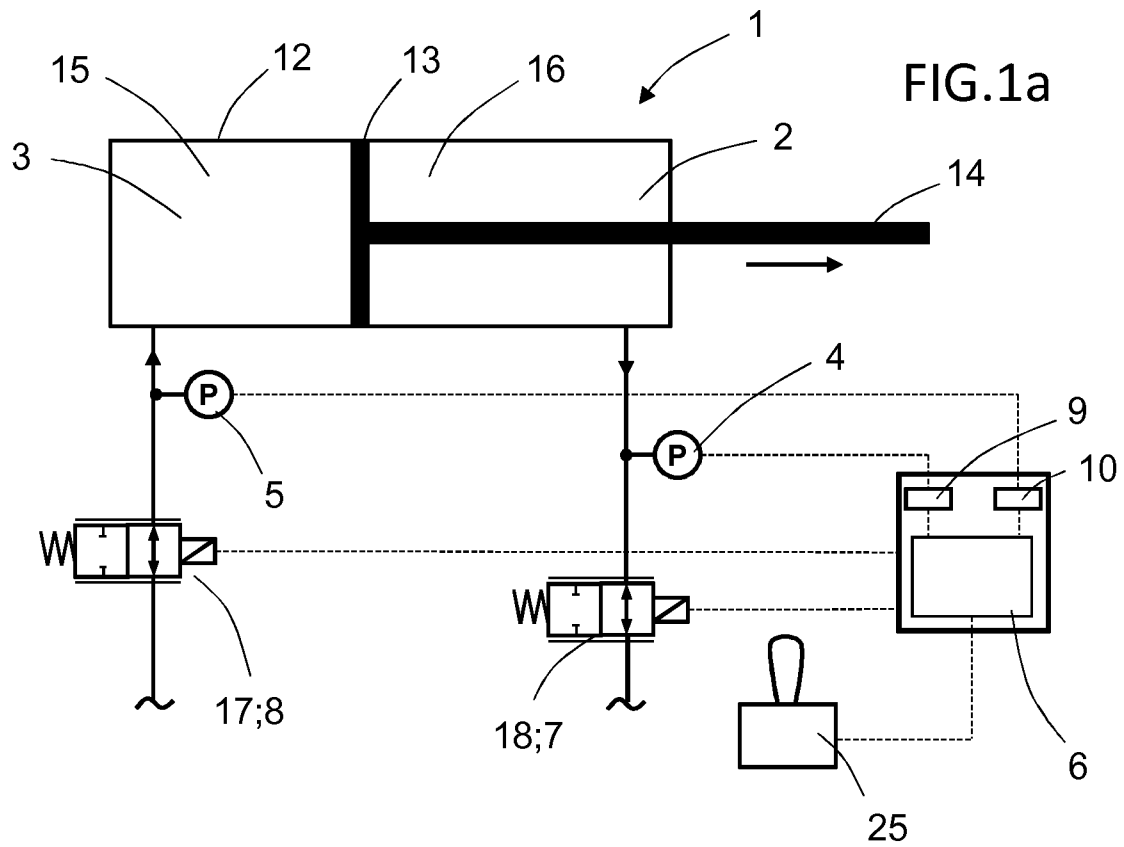
10 23. A system according to claim 20, wherein the third processing filter (20) comprises a high-pass filter with a cut-off frequency in the range of 0.1 – 10 Hz, specifically in the range of 0.5 – 5 Hz.

24. A system according to any of claims 21 to 23,
15 wherein the control signal from the control unit (6) is based on the sum of the low-pass filtered second pressure signal and the high-pass filtered first pressure signal, or
wherein the control signal from the control unit (6) is based on the sum of the low-pass filtered second pressure signal and the high-pass filtered second pressure signal and the high-pass filtered first pressure signal.
20

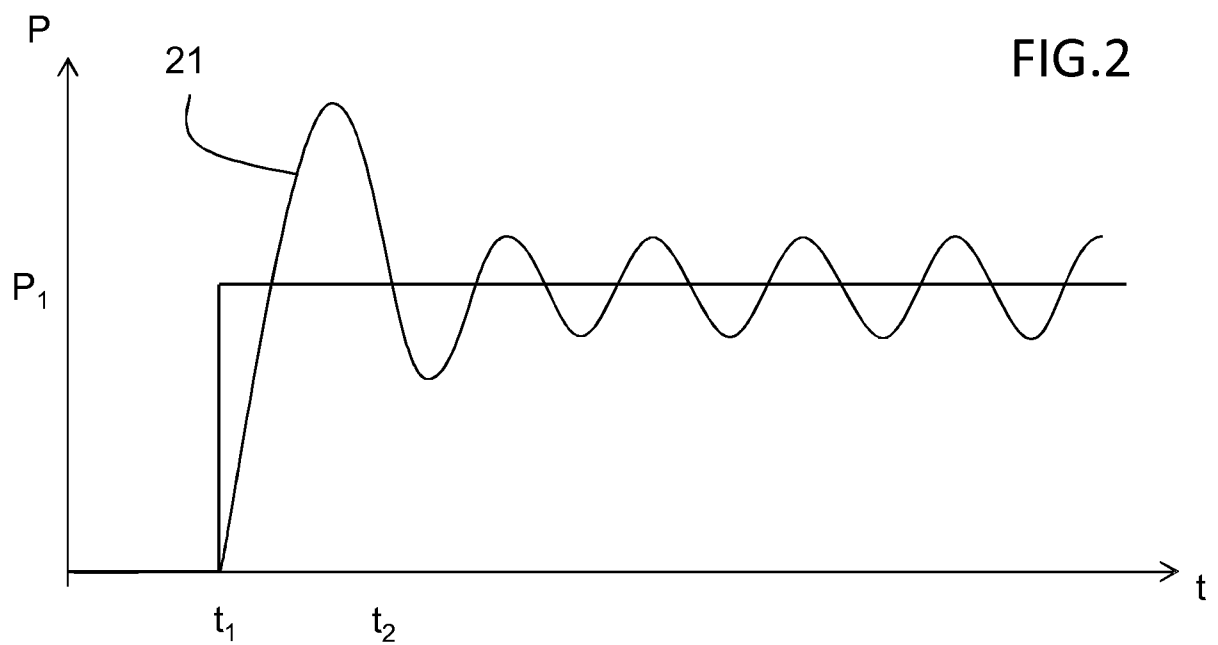
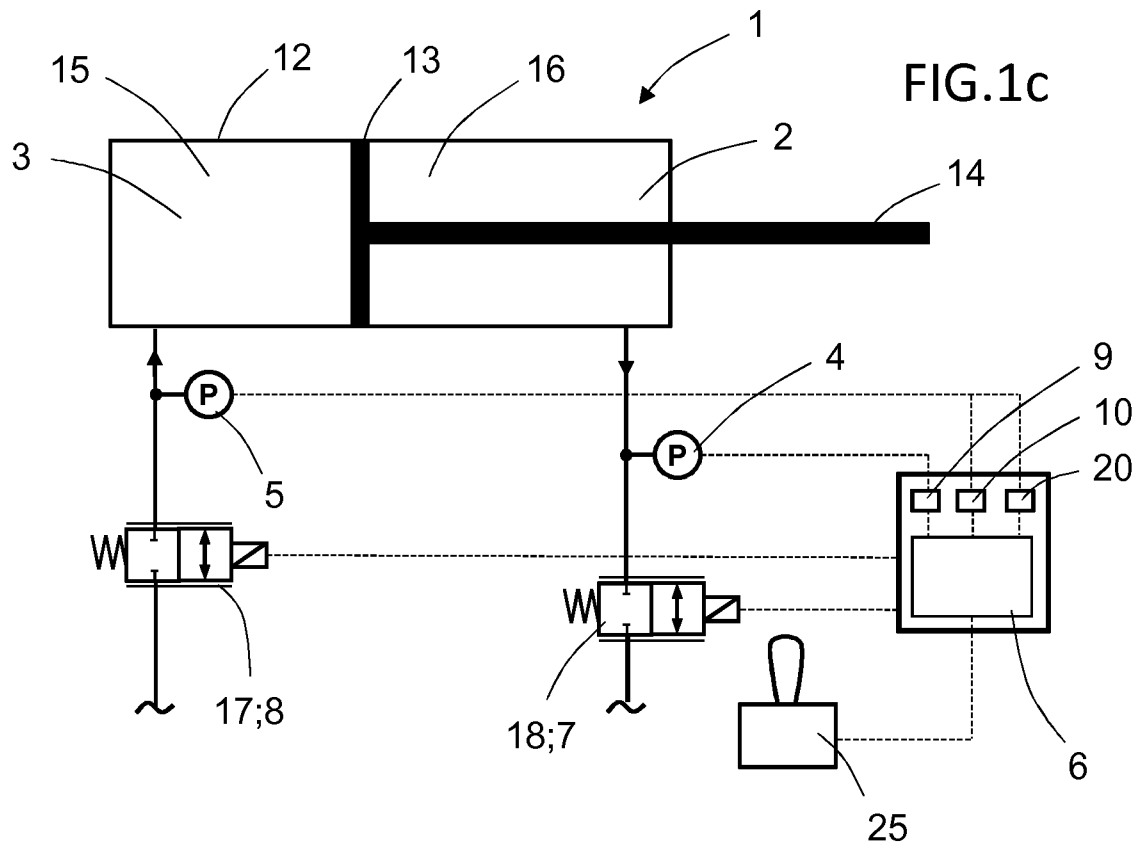
25 25. A system according to any of claims 16 to 23, wherein the control signal from the control unit (6) is based on the sum of the non-filtered second pressure signal and the high-pass filtered first pressure signal.

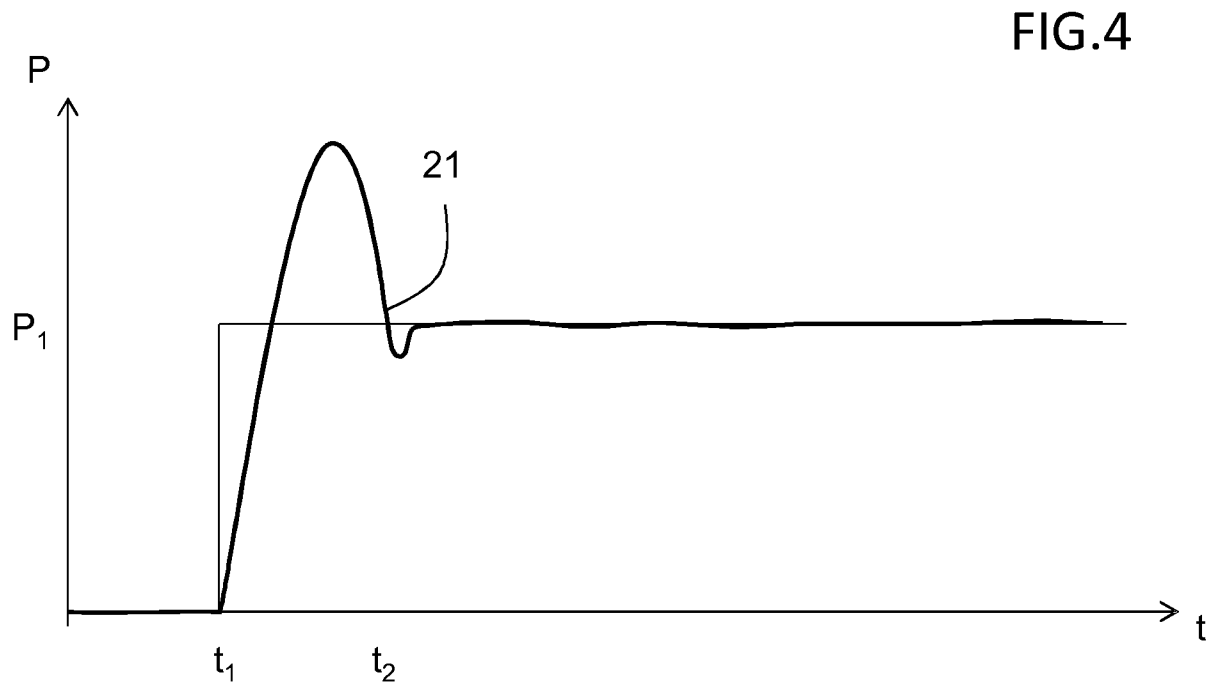
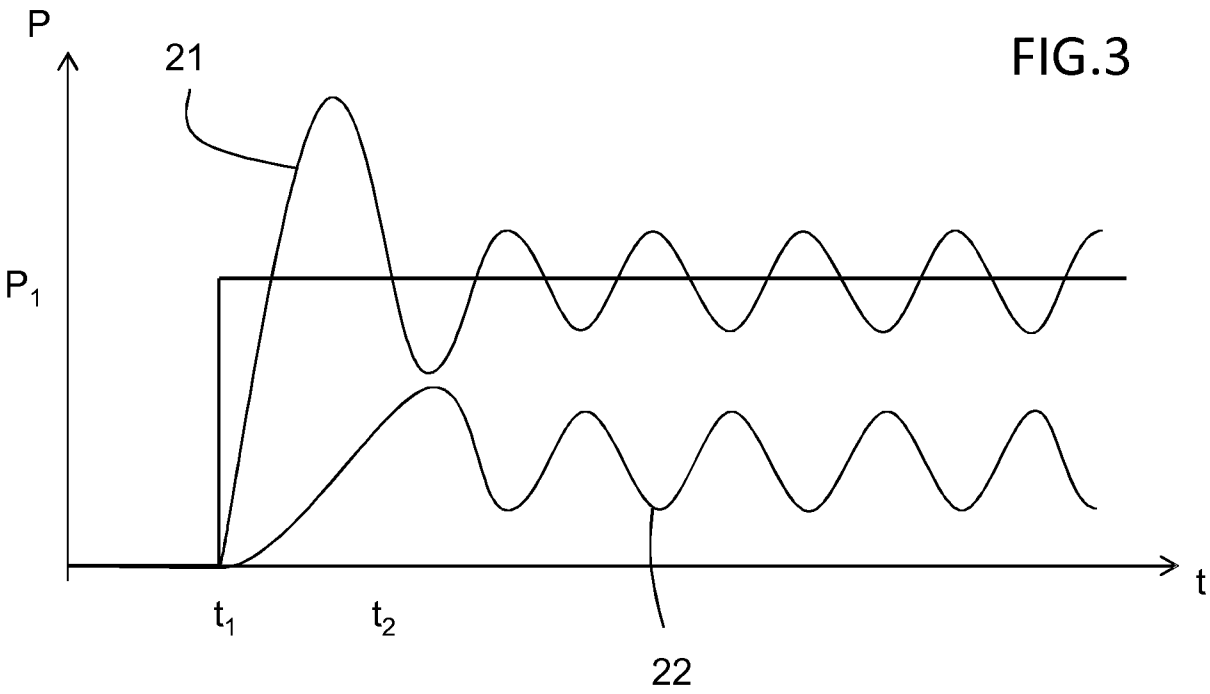
26. A system according to any of claims 14 to 25, wherein a valve position sensor is adapted for measuring the actuating position of the meter-in valve (8) and for generating a valve position signal, wherein the system is adapted to supply the valve position signal to the control unit (6), and
30 wherein the control unit (6) is adapted to generate a control signal based also on the valve position signal and to deliver the control signal to the meter-out valve (7).

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INTERNATIONAL SEARCH REPORT

International application No

PCT/EP2018/056109

A. CLASSIFICATION OF SUBJECT MATTER

INV. F15B11/042 F15B11/044 F15B20/00
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
F15B

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

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A	EP 2 811 174 A1 (DANFOSS POWER SOLUTIONS APS [DK]) 10 December 2014 (2014-12-10) figures 1,2 -----	1,14
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Further documents are listed in the continuation of Box C.



See patent family annex.

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"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

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INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

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